

Synchronized-Gap Rigidity and Exact Local Lattice Costs

A Non-Duplicative Progress Report on the Alaoglu–Erdős Conjecture

Report V

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Abstract

Assume hypothetically that a nonintegral real number β satisfies $M = 2^\beta \in \mathbb{Z}$ and $A = 3^\beta \in \mathbb{Z}$. This report develops a new frontier that is deliberately separate from the extensive reductions, p -adic determinant experiments, interpolation arguments, Sturmian codings, and auxiliary-function programs already present in the attached unified report. The principal global result proved here, conditional only on a standard theorem of Levin on generalized greatest common divisors in algebraic tori, is

$$\log \gcd(M^s - 2^r, A^s - 3^r) = o(r + s) \quad (r + s \rightarrow \infty).$$

Thus every synchronized pair of integer gaps has only a subexponential common divisor. An exceptional-subtorus descent is included; it is the step needed to make the statement uniform on the whole rank-two orbit, not merely away from an unspecified exceptional set.

The report then calculates the exact reduced height of the natural synchronized quotient and proves that its approximation exponent to $\log_2 3$ is asymptotically zero. This closes a tempting rational-approximation route. On the local side, a Smith-normal-form calculation gives the exact index of every p -adic exponent-congruence lattice. At a generic prime, p^k of additional principal-unit precision costs p^{2k} exponent classes. Minkowski’s theorem converts this into a precise “square-root tax”: ordinary local lifting consumes two lattice dimensions while the archimedean target has only one approximation coordinate. The resulting synthesis identifies the missing ingredient as an adelic rank-transfer phenomenon, not simply stronger pointwise lifting. Several quantitatively stated targets, an exact Fermat-quotient reconnaissance program, and a Lean formalization blueprint are supplied. No complete proof of the conjecture is claimed.

Status at a glance

New global theorem	Synchronized gaps have subexponential full gcd; the prime-to-6 part follows from toric generalized-gcd theory and the primes 2, 3 are controlled by p -adic logarithmic forms.
New exact calculation	The reduced quotient has height $\exp\{s \log(M_0 A_0) + o(s)\}$, so its convergence to $\log_2 3$ is far too weak for irrationality-measure arguments.
New local accounting	Smith invariants give the exact congruence-lattice index at every good prime and expose the generic quadratic cost.
Remaining bottleneck	A mechanism forcing coherent rank loss or a macroscopic aggregate determinant defect across finite places.
Proof status	Substantial conditional progress, but not a proof of the Alaoglu–Erdős conjecture.

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1 Scope, non-duplication, and public-status audit

1.1 What is and is not repeated

The attached unified report is unusually comprehensive. It already contains the standard irrationality reduction, the conditional monoid of integer values, primitive-valuation constructions, archimedean and p -adic determinant programs, Fermat-quotient layers, Kummer-theoretic reformulations, interpolation determinants, Subspace-Theorem barriers, Sturmian encodings, and many quantified failure analyses. Rephrasing those chapters would add volume but not progress.

This report therefore uses only the common starting notation and develops five pieces not presented there as a single rigorous mechanism:

1. a global generalized-gcd theorem for the two synchronized gaps, including descent through every exceptional torus coset;
2. an exact cancellation identity and height asymptotic for the natural quotient of those gaps;
3. a Smith-normal-form law for the full tower of local exponent congruences, not merely its first determinant test;
4. a geometry-of-numbers theorem translating local index into the best archimedean approximation one can force; and
5. an adelic defect budget that states exactly how much exceptional local behavior would be needed to overcome the generic square law.

The main purpose is diagnostic as well as constructive: each theorem eliminates a vague hope and replaces it with a numerical target.

1.2 Audit of the competitive claims

The public record as of August 23, 2026 contains a manuscript by Elkies and Klagsbrun constructing elliptic curves of rank at least 29 and 30, with machine-checked components reported in the accompanying material [10]. It also contains a public 2026 manuscript and seminar discussion proposing a counterexample to the Jacobian conjecture [11, 12]; a proposal of this scale should not be confused with settled community consensus until the proof has survived independent checking. The ICARM Hadamard project publicly records the completion of the formerly unresolved orders below 2000 and provides construction data [13].

I found no public preprint, code repository, or formal proof announcing a resolution of the Alaoglu–Erdős conjecture. The rival team’s private statement therefore cannot presently be audited. The correct response is not to infer a theorem from competitive pressure, but to make every new claim below independently checkable.

2 Conditional setup and synchronized gaps

Assume for the rest of the mathematical development that

$$\beta \in \mathbb{R} \setminus \mathbb{Z}, \quad M := 2^\beta \in \mathbb{Z}_{>0}, \quad A := 3^\beta \in \mathbb{Z}_{>0}. \quad (1)$$

A rational $\beta = u/v$ in lowest terms would make $2^{u/v}$ integral only when $v = 1$, so any counterexample is automatically irrational. In particular,

$$\frac{\log M}{\log 2} = \frac{\log A}{\log 3} = \beta, \quad \log M \log 3 - \log A \log 2 = 0. \quad (2)$$

Neither M is a power of 2 nor A a power of 3.

For integers $r, s \geq 1$ define the synchronized gaps

$$D_2(r, s) := M^s - 2^r, \quad D_3(r, s) := A^s - 3^r, \quad \delta_{r,s} := s\beta - r. \quad (3)$$

They satisfy the exact archimedean identities

$$D_2(r, s) = 2^r (2^{\delta_{r,s}} - 1), \quad D_3(r, s) = 3^r (3^{\delta_{r,s}} - 1). \quad (4)$$

When r/s approximates β , the two gaps are simultaneously small relative to their ambient powers. Their common divisors are therefore a natural place to look for a contradiction.

Two elementary facts will be used repeatedly.

Lemma 2.1 (Multiplicative independence). *Under (1), the pairs $(2, 3)$ and (M, A) are multiplicatively independent in the following strong sense: if*

$$2^u 3^v = 1 \quad \text{or} \quad M^u A^v = 1 \quad (u, v \in \mathbb{Z}),$$

then $u = v = 0$.

Proof. Unique factorization gives the assertion for $(2, 3)$. For (M, A) , take real logarithms and use (2): $u \log M + v \log A = \beta(u \log 2 + v \log 3) = 0$. As $\beta \neq 0$, unique factorization again gives $u = v = 0$. $\square$

Lemma 2.2 (Polynomial lower bound for a synchronized error). *There are effectively computable constants $C_0, C_1 > 0$, depending only on M , such that whenever $r, s \geq 1$ and $\delta_{r,s} \neq 0$,*

$$|\delta_{r,s}| \geq C_0 (\max\{r, s\})^{-C_1}. \quad (5)$$

Consequently, along any sequence $r/s \rightarrow \beta$,

$$\log |2^{\delta_{r,s}} - 1| = o(s), \quad \log |3^{\delta_{r,s}} - 1| = o(s). \quad (6)$$

Proof. The nonzero linear form $s \log M - r \log 2 = \delta_{r,s} \log 2$ is a fixed-base linear form in two logarithms. A standard Baker–Wüstholz estimate gives $|s \log M - r \log 2| \gg \max\{r, s\}^{-C_1}$, proving (5). If $\delta = o(s)$, an elementary upper bound gives $\log |b^\delta - 1| \leq O(|\delta| + 1) = o(s)$ away from zero, while (5) gives the lower bound $-O(\log s) = o(s)$ near zero. This proves (6) for $b = 2, 3$. $\square$

3 The rank-two torus orbit

Introduce the rational torus points

$$P_{r,s} := (M^s 2^{-r}, A^s 3^{-r}) \in \mathbb{G}_m^2(\mathbb{Q}) \quad (7)$$

and the finitely generated subgroup

$$\Gamma := \langle (2^{-1}, 3^{-1}), (M, A) \rangle \subset \mathbb{G}_m^2(\mathbb{Q}). \quad (8)$$

The map $(r, s) \mapsto P_{r,s}$ is injective: equality of two points would give $2^{r-r'} = M^{s-s'}$, and irrationality of β forces $r = r'$ and $s = s'$.

Proposition 3.1 (Zariski density). *The subgroup Γ is Zariski dense in $\mathbb{G}_m^2$.*

Proof. The Zariski closure of a subgroup of a torus is an algebraic subgroup. If it were proper, some nontrivial character $X^u Y^v$ would be trivial on its identity component and, after taking a positive power, on Γ . In particular, evaluating on $(2^{-1}, 3^{-1})$ would give $2^u 3^v = 1$. By Theorem 2.1, $u = v = 0$, a contradiction. Equivalently, no nontrivial character vanishes identically on Γ . $\square$

3.1 Generalized gcd on a torus

For rational numbers x, y , write the finite-place logarithmic gcd as

$$\text{lgcd}_{\text{fin}}(x, y) := \sum_p \min\{\max(0, v_p(x)), \max(0, v_p(y))\} \log p. \quad (9)$$

For algebraic numbers one sums over normalized nonarchimedean places; a full generalized gcd also includes archimedean proximity terms. We need only the consequence of Levin’s theorem for the coprime Laurent polynomials

$$f(X, Y) = X - 1, \quad g(X, Y) = Y - 1.$$

In the form used below, the theorem says that for every $\varepsilon > 0$, the generalized gcd of $f(P)$ and $g(P)$ is at most $\varepsilon h(P)$ for points P in a finitely generated subgroup, except on a finite union of translates of proper subtori [7, 8]. Discarding archimedean terms only weakens the estimate.

At a prime $p \geq 5$, the denominators 2^r and 3^r in (7) are p -adic units. Hence

$$v_p(M^s 2^{-r} - 1) = v_p(D_2(r, s)), \quad (10)$$

$$v_p(A^s 3^{-r} - 1) = v_p(D_3(r, s)). \quad (11)$$

Thus the prime-to-6 common divisor of the integer gaps is exactly the finite local gcd of the torus coordinates at the good primes.

4 Global synchronized-gap rigidity

A direct invocation of the toric theorem would leave an exceptional union of torus cosets. That is not enough: a hypothetical counterexample could conceivably force every useful approximant into those cosets. The next argument descends through the exceptional set and is the main global advance of this report.

4.1 Pullback of exceptional cosets

Let $C = \xi H$ be a translate of a proper subtorus $H \subset \mathbb{G}_m^2$. If $C \cap \Gamma$ is nonempty, choose $\gamma_0 \in C \cap \Gamma$. Then

$$C \cap \Gamma = \gamma_0(\Gamma \cap H). \quad (12)$$

Because $\Gamma \cong \mathbb{Z}^2$ and is Zariski dense, $\Gamma \cap H$ has rank at most one. Under the injective exponent parametrization, the pullback of $C \cap \Gamma$ is therefore either empty, a single point, or an affine rank-one lattice

$$(r, s) = (r_0, s_0) + t(a, b), \quad t \in \mathbb{Z}, \quad (13)$$

for a nonzero primitive vector $(a, b) \in \mathbb{Z}^2$.

Along such a line,

$$P_{r,s} = (c_1 \rho^t, c_2 \sigma^t), \quad (14)$$

where

$$c_1 = M^{s_0} 2^{-r_0}, \quad c_2 = A^{s_0} 3^{-r_0}, \quad \rho = M^b 2^{-a}, \quad \sigma = A^b 3^{-a}. \quad (15)$$

Lemma 4.1 (Every exceptional line has a dense cyclic direction). *For every nonzero $(a, b) \in \mathbb{Z}^2$, the numbers ρ and σ in (15) are multiplicatively independent. Hence $\{(\rho^t, \sigma^t) : t \in \mathbb{Z}\}$ is Zariski dense in $\mathbb{G}_m^2$.*

Proof. Suppose $\rho^u \sigma^v = 1$. Taking real logarithms and using (2) gives

$$\begin{aligned} 0 &= u(b \log M - a \log 2) + v(b \log A - a \log 3) \\ &= (b\beta - a)(u \log 2 + v \log 3). \end{aligned}$$

Irrationality of β and nonvanishing of (a, b) imply $b\beta - a \neq 0$. Therefore $2^u 3^v = 1$, so $u = v = 0$. A cyclic subgroup of $\mathbb{G}_m^2$ is Zariski dense precisely when its two coordinates are multiplicatively independent. $\square$

4.2 The prime-to-6 theorem

For a nonzero integer n , let

$$n_{(6)} := \frac{|n|}{2^{v_2(n)} 3^{v_3(n)}}, \quad \gcd_{(6)}(u, v) := \gcd(u_{(6)}, v_{(6)}). \quad (16)$$

Theorem 4.2 (Subexponential prime-to-6 synchronized gcd). *Under (1),*

$$\log \gcd_{(6)}(D_2(r, s), D_3(r, s)) = o(r + s) \quad (17)$$

as $r, s \geq 1$ and $r + s \rightarrow \infty$. Equivalently, for every $\varepsilon > 0$ there is $H_0(\varepsilon, M, A)$ such that

$$\gcd_{(6)}(D_2(r, s), D_3(r, s)) \leq \exp(\varepsilon(r + s)) \quad (18)$$

whenever $r + s \geq H_0$.

Proof. Fix $\varepsilon > 0$. Apply Levin’s toric generalized-gcd theorem to the Zariski-dense group Γ and the coprime functions $X - 1, Y - 1$. Outside a finite union Z_ε of translates of proper subtori,

$$\lg \gcd_{\text{fin}}(P_1 - 1, P_2 - 1) \leq \varepsilon' h(P) \quad (19)$$

for an $\varepsilon' > 0$ chosen below. The logarithmic height of $P_{r,s}$ is $O(r + s)$, with a constant depending only on M, A . By (10)–(11), the contribution from $p \geq 5$ is exactly $\log \gcd_{(6)}(D_2, D_3)$. Choosing ε' small enough therefore gives (18) outside Z_ε .

It remains to handle $P_{r,s} \in Z_\varepsilon$. By (12), the exponent pairs in this intersection lie in finitely many points and affine lines of the form (13). Isolated points are irrelevant asymptotically. Fix one line. The points have the cyclic form (14), and Theorem 4.1 says that its direction (ρ, σ) is Zariski dense.

Apply the same generalized-gcd theorem to the finitely generated subgroup generated by (c_1, c_2) and (ρ, σ) . It gives a second finite exceptional union of proper torus cosets. Each such coset meets the sequence $(c_1 \rho^t, c_2 \sigma^t)$ only finitely often: if it contained two terms with parameters

$t_1 \neq t_2$, then $(\rho, \sigma)^{t_1-t_2}$ would lie in a proper subtorus, contradicting the Zariski density of its cyclic powers. Thus, after finitely many values of t , the generalized gcd on this line is $o(|t|)$.

There are only finitely many first-stage exceptional lines. Taking the maximum of their finite thresholds and rescaling ε' by the finitely many height constants yields a uniform $o(r+s)$ estimate on all of them. Combining the generic region, the exceptional lines, and the isolated points proves the theorem. $\square$

Remark 4.3 (Why the second descent matters). The conclusion is not a routine restatement of “gcd is small outside an exceptional set.” The first exceptional set depends on ε and may contain infinitely many exponent pairs. The cyclic-density argument shows that every rank-one component carries a fresh Zariski-dense orbit, so the exceptional mechanism cannot persist after one descent. In a two-dimensional torus, the descent terminates.

4.3 Restoring the structural primes

The denominator choices in (7) single out 2 and 3. They are not controlled by the preceding toric identity, but they are much easier.

Proposition 4.4 (Structural-prime contribution). *There is a constant $C = C(M, A) > 0$ such that*

$$\min\{v_2(D_2), v_2(D_3)\} + \min\{v_3(D_2), v_3(D_3)\} \leq C(1 + \log(r+s))^2. \quad (20)$$

Proof. For the prime 2, the common valuation is at most $v_2(A^s - 3^r)$. If A is even, this difference is odd and the valuation is zero. If A is odd, then A and 3 are multiplicatively independent; otherwise $A = 3^m$ and (2) would give $\beta = m \in \mathbb{Z}$. A fixed-base 2-adic linear-form estimate therefore gives $v_2(A^s - 3^r) \ll (1 + \log(r+s))^2$.

Similarly, the common 3-adic valuation is at most $v_3(M^s - 2^r)$. It is zero when $3 \mid M$, and otherwise the multiplicative independence of M and 2, followed by a fixed-base 3-adic logarithmic-form estimate, gives the same polylogarithmic bound. Standard results of Yu or of Bugeaud–Laurent are more than sufficient here [9, 3]. $\square$

Theorem 4.5 (Full synchronized-gap rigidity). *Under (1),*

$$\boxed{\log \gcd(M^s - 2^r, A^s - 3^r) = o(r+s)}. \quad (21)$$

Proof. The prime-to-6 contribution is $o(r+s)$ by Theorem 4.2; the two remaining local contributions are $O((\log(r+s))^2) = o(r+s)$ by Theorem 4.4. $\square$

Interpretation of the global theorem

No sequence of synchronized gaps can carry a common divisor of size $\exp(c(r+s))$ for a fixed $c > 0$. Any proof strategy that tries to force many local congruences simultaneously must therefore pay enough exponent height to keep the total forced divisor subexponential. The theorem does not itself contradict the existence of β ; it supplies a global conservation law against which every local construction can be measured.

5 Exact cancellation and the height of the synchronized quotient

A natural reaction to (4) is to divide the two small gaps. If $\delta_{r,s} \rightarrow 0$, then

$$R_{r,s} := \frac{A^s/3^r - 1}{M^s/2^r - 1} = \frac{3^{\delta_{r,s}} - 1}{2^{\delta_{r,s}} - 1} \longrightarrow \frac{\log 3}{\log 2}. \quad (22)$$

Since $R_{r,s} \in \mathbb{Q}$, one might hope for rational approximations too good for the known irrationality measure of $\log_2 3$. The following exact height calculation shows why that hope fails.

Write

$$a := v_2(M), \quad b := v_3(A), \quad M_0 := M/2^a, \quad A_0 := A/3^b. \quad (23)$$

Because β is not an integer, $M_0 > 1$ and $A_0 > 1$. Indeed, $M_0 = 1$ would make $M = 2^a$ and hence $\beta = a$; similarly for A_0 . Moreover

$$\beta = a + \log_2 M_0 = b + \log_3 A_0, \quad (24)$$

so $\beta > a, b$.

For any sequence $s \rightarrow \infty$ with $r/s \rightarrow \beta$, eventually $r > as$ and $r > bs$. Define

$$E_2(r, s) := M_0^s - 2^{r-as}, \quad E_3(r, s) := A_0^s - 3^{r-bs}. \quad (25)$$

Then E_2 is odd and $3 \nmid E_3$, while

$$D_2 = 2^{as} E_2, \quad D_3 = 3^{bs} E_3. \quad (26)$$

Proposition 5.1 (Exact reduction identity). *For $r > as, bs$, one has*

$$R_{r,s} = \frac{2^{r-as} E_3(r, s)}{3^{r-bs} E_2(r, s)}. \quad (27)$$

If $g_{r,s} := \gcd_{(6)}(D_2, D_3)$, then the numerator and denominator on the right of (27) have greatest common divisor exactly $g_{r,s}$. Consequently the reduced numerator and denominator are

$$P_{r,s} = \frac{2^{r-as} |E_3|}{g_{r,s}}, \quad Q_{r,s} = \frac{3^{r-bs} |E_2|}{g_{r,s}}. \quad (28)$$

Proof. Starting with (22),

$$R_{r,s} = \frac{2^r (A^s - 3^r)}{3^r (M^s - 2^r)} = \frac{2^{r-as} E_3}{3^{r-bs} E_2},$$

which proves (27). Since E_2 is odd, the denominator has no factor 2 in common with the numerator. Since $3 \nmid E_3$, there is no common factor 3. At every prime $p \geq 5$, the common valuation is $\min\{v_p(E_2), v_p(E_3)\} = \min\{v_p(D_2), v_p(D_3)\}$. This is exactly $g_{r,s}$. $\square$

Theorem 5.2 (Exact logarithmic height asymptotic). *Let $s \rightarrow \infty$ and $r/s \rightarrow \beta$. In lowest terms, the rational number $R_{r,s}$ has multiplicative height*

$$\boxed{\log H(R_{r,s}) = s \log(M_0 A_0) + o(s)}. \quad (29)$$

More precisely,

$$\log P_{r,s} = s \log(M_0 A_0) + o(s), \quad \log Q_{r,s} = s \log(M_0 A_0) + o(s). \quad (30)$$

Proof. Using $M_0^s = 2^{s(\beta-a)}$ and $A_0^s = 3^{s(\beta-b)}$,

$$E_2 = 2^{r-as}(2^{\delta_{r,s}} - 1), \quad (31)$$

$$E_3 = 3^{r-bs}(3^{\delta_{r,s}} - 1). \quad (32)$$

By Theorem 2.2, the logarithms of the final factors in (31)–(32) are $o(s)$. Since $r/s \rightarrow \beta$,

$$\log |E_2| = (r - as) \log 2 + o(s) = s \log M_0 + o(s), \quad (33)$$

$$\log |E_3| = (r - bs) \log 3 + o(s) = s \log A_0 + o(s). \quad (34)$$

The global theorem Theorem 4.2 gives $\log g_{r,s} = o(r + s) = o(s)$. Substitution into (28) yields

$$\begin{aligned} \log P_{r,s} &= (r - as) \log 2 + \log |E_3| - \log g_{r,s} \\ &= s \log M_0 + s \log A_0 + o(s), \end{aligned}$$

and the same calculation gives the formula for $Q_{r,s}$. □

5.1 The rational-approximation route is quantitatively closed

Set

$$F(t) := \frac{3^t - 1}{2^t - 1}, \quad \theta := \frac{\log 3}{\log 2}. \quad (35)$$

The removable singularity of F at 0 has expansion

$$F(t) = \theta + \frac{\theta(\log 3 - \log 2)}{2} t + O(t^2). \quad (36)$$

Corollary 5.3 (Zero power-of-height exponent). *Along every sequence with $r/s \rightarrow \beta$ and $\delta_{r,s} \rightarrow 0$,*

$$\frac{-\log |R_{r,s} - \theta|}{\log H(R_{r,s})} \rightarrow 0. \quad (37)$$

In particular, for every fixed $\mu > 0$ and all sufficiently large terms,

$$|R_{r,s} - \theta| > H(R_{r,s})^{-\mu}. \quad (38)$$

Proof. The coefficient of t in (36) is nonzero. Therefore, for sufficiently small nonzero t , $|F(t) - \theta| \asymp |t|$. The Baker lower bound (5) implies $-\log |\delta_{r,s}| = O(\log s)$. On the other hand, Theorem 5.2 gives $\log H(R_{r,s}) = s \log(M_0 A_0) + o(s)$, with a positive leading constant. Their ratio tends to zero. □

A closed door, not merely a failed estimate

The quotient $R_{r,s}$ does converge to $\log_2 3$ whenever $s\beta - r \rightarrow 0$, but its numerator and denominator each retain essentially all of the primitive exponential mass $M_0^s A_0^s$. The common-gap gcd is too small to cancel that mass. Hence the approximation is logarithmic in its height, not a positive power of its height. Improvements to the irrationality measure of $\log_2 3$ cannot rescue this particular route.

6 Exact local congruence lattices

The global theorem says that common divisibility cannot be too cheap in aggregate. We now calculate the local price exactly. This section uses only elementary p -adic Lie theory and Smith normal form.

Fix a prime

$$p \geq 5, \quad p \nmid 6MA. \quad (39)$$

For a p -adic unit u , write

$$u = \omega_p(u) \langle u \rangle_p, \quad \omega_p(u)^{p-1} = 1, \quad \langle u \rangle_p \in 1 + p\mathbb{Z}_p. \quad (40)$$

The Iwasawa logarithm kills the Teichmüller factor and satisfies $\log_p u \in p\mathbb{Z}_p$. Define the normalized logarithm matrix

$$B_p := \frac{1}{p} \begin{pmatrix} -\log_p 2 & \log_p M \\ -\log_p 3 & \log_p A \end{pmatrix} \in M_2(\mathbb{Z}_p). \quad (41)$$

For $z = (r, s)^\top$,

$$B_p z = \frac{1}{p} \begin{pmatrix} s \log_p M - r \log_p 2 \\ s \log_p A - r \log_p 3 \end{pmatrix}. \quad (42)$$

Definition 6.1 (Principal-unit exponent lattice). For $k \geq 0$, set

$$\Lambda_p(k) := \{z \in \mathbb{Z}^2 : B_p z \in p^k \mathbb{Z}_p^2\}. \quad (43)$$

Membership is equivalent to the two principal-unit congruences

$$\langle M \rangle_p^s \langle 2 \rangle_p^{-r} \equiv 1 \pmod{p^{k+1}}, \quad \langle A \rangle_p^s \langle 3 \rangle_p^{-r} \equiv 1 \pmod{p^{k+1}}. \quad (44)$$

6.1 The Smith-index law

Let $x_+ := \max(x, 0)$. If B_p has rank two, there are $U, V \in \mathrm{GL}_2(\mathbb{Z}_p)$ and uniquely determined integers $0 \leq \alpha_p \leq \gamma_p$ such that

$$UB_p V = \mathrm{diag}(p^{\alpha_p}, p^{\gamma_p}). \quad (45)$$

Then

$$\tau_p := v_p(\det B_p) = \alpha_p + \gamma_p. \quad (46)$$

For rank one, write the Smith form as $\mathrm{diag}(p^{\alpha_p}, 0)$.

Theorem 6.2 (Exact local index). *For every $k \geq 0$:*

1. *if $\mathrm{rank} B_p = 2$, then*

$$[\mathbb{Z}^2 : \Lambda_p(k)] = p^{(k-\alpha_p)_+ + (k-\gamma_p)_+}; \quad (47)$$

in particular, for $k \geq \gamma_p$,

$$[\mathbb{Z}^2 : \Lambda_p(k)] = p^{2k-\tau_p}; \quad (48)$$

2. *if $\mathrm{rank} B_p = 1$ with nonzero invariant p^{α_p} , then*

$$[\mathbb{Z}^2 : \Lambda_p(k)] = p^{(k-\alpha_p)_+}; \quad (49)$$

3. *if $B_p = 0$, then $\Lambda_p(k) = \mathbb{Z}^2$.*

Proof. Complete $\mathbb{Z}^2$ at p . Since the defining condition has finite p -power index, its index in $\mathbb{Z}^2$ equals the index of

$$\{z \in \mathbb{Z}_p^2 : B_p z \in p^k \mathbb{Z}_p^2\}$$

in $\mathbb{Z}_p^2$. Multiplication by U and V in (45) preserves Haar measure and index. For a diagonal entry p^e , the condition $p^e x \in p^k \mathbb{Z}_p$ imposes $x \in p^{(k-e)+} \mathbb{Z}_p$ and therefore contributes $p^{(k-e)+}$. Multiplying the independent coordinate contributions gives (47) and (49). A zero diagonal entry imposes no condition. $\square$

Corollary 6.3 (Generic quadratic law). *If $\det B_p \in \mathbb{Z}_p^\times$, then $\alpha_p = \gamma_p = 0$ and*

$$[\mathbb{Z}^2 : \Lambda_p(k)] = p^{2k} \quad (k \geq 0). \quad (50)$$

Thus p^k of additional principal-unit precision costs p^{2k} exponent classes.

Remark 6.4 (Defect versus rank collapse). A finite determinant valuation $\tau_p > 0$ gives only a fixed credit p^{τ_p} once $k \geq \gamma_p$; the slope in k remains 2. A true rank-one identity $\det B_p = 0$ changes the slope from 2 to 1. This is why a first-layer vanishing congruence is not by itself the phenomenon a proof needs. One must distinguish

$$\det B_p \equiv 0 \pmod{p}, \quad v_p(\det B_p) \gg 1, \quad \det B_p = 0 \text{ in } \mathbb{Q}_p.$$

They have qualitatively different asymptotic effects.

6.2 First-layer Fermat quotient

For $p \nmid u$, define

$$q_p(u) := \frac{u^{p-1} - 1}{p} \pmod{p}. \quad (51)$$

The standard congruence

$$\frac{\log_p u}{p} \equiv -q_p(u) \pmod{p} \quad (52)$$

turns the determinant of (41) into a completely elementary first-layer test.

Proposition 6.5 (Fermat-quotient shadow). *Define*

$$\Delta_p(M, A) := q_p(M)q_p(3) - q_p(A)q_p(2) \in \mathbb{F}_p. \quad (53)$$

Then

$$\det B_p \equiv \Delta_p(M, A) \pmod{p}. \quad (54)$$

Consequently, $\Delta_p(M, A) \neq 0$ implies the exact generic law $[\mathbb{Z}^2 : \Lambda_p(k)] = p^{2k}$ for all k .

Proof. Modulo p , the four entries of B_p are respectively $q_p(2), -q_p(M), q_p(3), -q_p(A)$. Taking the determinant gives $q_p(M)q_p(3) - q_p(A)q_p(2)$. $\square$

6.3 Restoring the Teichmüller layer

Principal-unit conditions do not by themselves imply the original integer congruences. Define the residue or torsion lattice

$$T_p := \{(r, s) \in \mathbb{Z}^2 : M^s \equiv 2^r \pmod{p}, \quad A^s \equiv 3^r \pmod{p}\}. \quad (55)$$

Let

$$t_p := [\mathbb{Z}^2 : T_p]. \quad (56)$$

The homomorphism

$$(r, s) \mapsto (M^s 2^{-r}, A^s 3^{-r}) \quad \text{in } (\mathbb{F}_p^\times)^2$$

has kernel T_p , so

$$t_p \mid (p-1)^2. \quad (57)$$

In particular, $p \nmid t_p$.

Theorem 6.6 (Full local congruence index). *The exponent pairs satisfying*

$$M^s \equiv 2^r \pmod{p^{k+1}}, \quad A^s \equiv 3^r \pmod{p^{k+1}} \quad (58)$$

form the lattice

$$C_p(k) = T_p \cap \Lambda_p(k), \quad (59)$$

and

$$[\mathbb{Z}^2 : C_p(k)] = t_p [\mathbb{Z}^2 : \Lambda_p(k)]. \quad (60)$$

Proof. The Teichmüller decomposition (40) separates the mod- p condition from the principal-unit condition, proving (59). The indices of T_p and $\Lambda_p(k)$ are coprime: the first divides $(p-1)^2$ and the second is a power of p . For finite-index subgroups of an abelian group with coprime indices, the index of the intersection is the product. $\square$

6.4 Several primes and the defect budget

For a finite set S of good primes and depths $\mathbf{k} = (k_p)_{p \in S}$, set

$$\Lambda_S(\mathbf{k}) := \bigcap_{p \in S} \Lambda_p(k_p), \quad G_S(\mathbf{k}) := \prod_{p \in S} p^{k_p}. \quad (61)$$

Because the individual indices are powers of distinct primes, the Chinese remainder theorem gives exact multiplication.

Theorem 6.7 (Exact finite-place product law). *One has*

$$[\mathbb{Z}^2 : \Lambda_S(\mathbf{k})] = \prod_{p \in S} [\mathbb{Z}^2 : \Lambda_p(k_p)]. \quad (62)$$

If every B_p has rank two and $k_p \geq \gamma_p$, then

$$[\mathbb{Z}^2 : \Lambda_S(\mathbf{k})] = \prod_{p \in S} p^{2k_p - \tau_p} = \frac{G_S(\mathbf{k})^2}{E_S}, \quad (63)$$

where the aggregate determinant credit is

$$E_S := \prod_{p \in S} p^{\tau_p}. \quad (64)$$

Proof. For each p , the quotient $\mathbb{Z}^2 / \Lambda_p(k_p)$ is a finite p -group. The natural map from $\mathbb{Z}^2$ to their product is surjective by the Chinese remainder theorem applied to a common modulus, and its kernel is the intersection. This proves (62); the stable Smith formula (48) gives (63). $\square$

Remark 6.8 (Torsion factors across several primes). For the full lattices $C_p(k_p)$, each local index is still given exactly by (60). Across distinct primes, however, the prime-to- p torsion factors t_p need not be pairwise coprime, so their intersection index is not automatically the product of all t_p . This bounded first layer is arithmetically important for constructing actual gap divisibility, but it does not change the asymptotic p -power slope that drives the square-root barrier.

7 Covolume conservation and the square-root tax

The Smith law is local. To use it, one must find a point in the resulting exponent lattice whose slope r/s is close to β . This is a geometry-of-numbers problem with an exact answer.

Theorem 7.1 (Constrained Dirichlet theorem). *Let $\Lambda \subset \mathbb{Z}^2$ be a lattice of finite index I , let $\beta \in \mathbb{R} \setminus \mathbb{Q}$, and let $Q > 1$, $\eta > 0$. There exists a nonzero $(r, s) \in \Lambda$ such that, after changing both signs if necessary,*

$$0 < s \leq (1 + \eta)IQ, \quad |s\beta - r| \leq Q^{-1}. \quad (65)$$

Proof. Consider the determinant-one linear map

$$T_\beta(r, s) := (s, s\beta - r). \quad (66)$$

The lattice $L = T_\beta(\Lambda)$ has covolume I . Let

$$K = [-X, X] \times [-Y, Y], \quad X = (1 + \eta)IQ, \quad Y = Q^{-1}.$$

Its area is $4XY = 4(1 + \eta)I > 4 \operatorname{covol}(L)$. Minkowski's convex-body theorem produces a nonzero point $T_\beta(r, s) \in L \cap K$. If $s = 0$, then $|r| = |s\beta - r| \leq Q^{-1} < 1$, forcing $r = 0$, contrary to nonzero-ness. Thus $s \neq 0$; replacing (r, s) by $(-r, -s)$ gives $s > 0$. $\square$

Corollary 7.2 (Height form). *For every $H > (1 + \eta)I$, the lattice Λ contains a nonzero (r, s) with $0 < s \leq H$ and*

$$|s\beta - r| \leq \frac{(1 + \eta)I}{H}. \quad (67)$$

Proof. Use Theorem 7.1 with $Q = H/((1 + \eta)I)$, decreasing H infinitesimally if necessary to retain the strict Minkowski inequality. $\square$

7.1 Generic local precision costs a square

Apply Theorem 7.2 to the principal-unit lattice $\Lambda_S(\mathbf{k})$. In the stable rank-two regime, (63) gives

$$I = \frac{G^2}{E}, \quad G := G_S(\mathbf{k}), \quad E := E_S. \quad (68)$$

Consequently, at exponent height H one can force both local logarithmic conditions and guarantee only

$$|s\beta - r| \lesssim \frac{G^2}{EH}. \quad (69)$$

If every selected prime is generic, $E = 1$ and the right side is G^2/H .

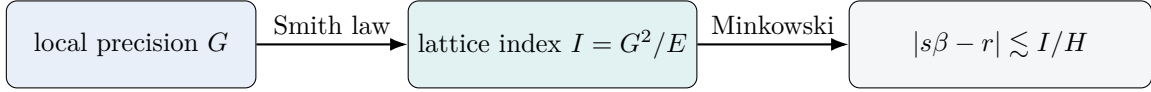
This is the square-root tax. To prescribe one p -adic coordinate to precision p^k costs p^k classes; prescribing two independent coordinates costs p^{2k} . Yet the archimedean approximation asks only for the one linear form $s\beta - r$ to be small. The second local condition consumes a full lattice dimension without adding a second archimedean degree of freedom.

Proposition 7.3 (Defect needed to recover linear cost). *In the full-rank stable regime, obtaining an essentially linear lattice cost $I \leq G^{1+o(1)}$ is equivalent to the aggregate condition*

$$\log E \geq (1 - o(1)) \log G, \quad (70)$$

or explicitly

$$\sum_{p \in S} \tau_p \log p \geq (1 - o(1)) \sum_{p \in S} k_p \log p. \quad (71)$$



Generic rank two gives $E = 1$: two local constraints create a quadratic cost before the one-dimensional real approximation begins.

Figure 1: The exact local-to-archimedean accounting chain.

Proof. From (68), $\log I = 2 \log G - \log E$. The assertion $\log I \leq (1 + o(1)) \log G$ is therefore exactly (70). $\square$

Remark 7.4 (Exact rank one). If B_p has rank one, then a depth k costs only $p^{k-\alpha_p}$ by (49). At such a prime the quadratic law genuinely collapses to a linear law. A family of rank-one places would therefore be qualitatively different from a family of isolated Fermat-quotient zeros with bounded τ_p . As Section 11 explains, even this slope change is not sufficient without exceptional archimedean coherence of the surviving directions.

7.2 Why the volume calculation is the right benchmark

The proof of Theorem 7.1 is elementary, but its interpretive force is substantial. The transformed lattice has covolume I , while the target rectangle has side scales H and $|s\beta - r|$. The area product must cross the covolume scale before Minkowski can guarantee a nonzero point. Thus any strategy that uses only finite-index information and a symmetric convex-body argument will inherit

$$H \cdot |s\beta - r| \asymp I. \quad (72)$$

Escaping the square law therefore requires additional arithmetic structure inside the lattice: unusually short directions, correlations among places, or a genuine rank collapse. A more elaborate convex body cannot create such structure by itself.

8 Global–local synthesis

The global theorem and the local index theorem describe the same tension from opposite sides.

8.1 Forced divisibility versus exponent height

Suppose (r, s) satisfies the full congruences

$$M^s \equiv 2^r \pmod{N}, \quad A^s \equiv 3^r \pmod{N}, \quad (73)$$

where N is supported on good primes. Then

$$N \mid \gcd(D_2(r, s), D_3(r, s)). \quad (74)$$

The global theorem gives, along any sequence with $r + s \rightarrow \infty$,

$$\log N = o(r + s). \quad (75)$$

Thus an exponentially large forced modulus cannot occur at linearly comparable exponent height.

On the other hand, the generic local law predicts that a principal-unit modulus of scale G has lattice index G^2 . Minkowski then naturally finds exponents of size at least comparable to G^2 before obtaining a fixed-quality real approximation. At that height, $\log G$ is only $O(\log(r+s))$, so (75) is easily respected. There is no contradiction because the generic local construction is far more expensive than the global theorem forbids.

The precise mismatch

The global theorem rules out $N = \exp(cH)$ at exponent height H . Generic local lifting produces only N of polynomial size in H because the congruence lattice has index roughly N^2 . Between polynomial and exponential lies a vast gap. A proof needs a mechanism that changes the local cost exponent, not a modest improvement in constants.

8.2 A phase diagram

Let G denote additional principal-unit precision, E the determinant credit, $I = G^2/E$ the exponent-lattice index, and H the selected exponent height. The following regimes are qualitatively distinct.

Local regime	Index scale	Minkowski error at height H	Assessment
Generic rank two	$I = G^2$	$ \delta \lesssim G^2/H$	Too expensive; no tension with the global gcd theorem.
Finite defects	$I = G^2/E, E = G^{o(1)}$	$ \delta \lesssim G^{2-o(1)}/H$	Same asymptotic slope; bounded-depth anomalies do not help.
Macroscopic defect	$E = G^{1-o(1)}$	$ \delta \lesssim G^{1+o(1)}/H$	Removes one dimension of index cost; direction compression is still required.
Rank one at many places	$I \approx G$	$ \delta \lesssim G/H$	A necessary structural shift, but insufficient without cross-prime short-direction coherence.
Rank zero	I bounded	$ \delta \lesssim H^{-1}$	Would mean both p -adic logarithmic relations vanish identically; extraordinarily rigid.

8.3 The real determinant and its finite-place shadows

The counterexample relation is the vanishing of the real determinant

$$D_\infty := \det \begin{pmatrix} \log 2 & \log M \\ \log 3 & \log A \end{pmatrix} = \log 2 \log A - \log 3 \log M = 0. \quad (76)$$

At a good finite prime define

$$D_p := \det \begin{pmatrix} \log_p 2 & \log_p M \\ \log_p 3 & \log_p A \end{pmatrix} = -p^2 \det B_p. \quad (77)$$

The conjecture asks, in effect, whether the real rank loss can occur without a rational rank loss. The local analysis shows that a proof would follow from a sufficiently strong theorem transferring the archimedean rank loss into aggregate finite-place rank loss.

There is no elementary product formula for the numbers D_v : each is a bilinear expression in logarithms, not the valuation of a single algebraic number. This is exactly why the missing theorem is nontrivial. Still, (76) and (77) organize the problem into an adelic wedge:

$$(\log_v 2, \log_v 3) \wedge (\log_v M, \log_v A), \quad v \leq \infty. \quad (78)$$

The archimedean wedge vanishes. A decisive principle would force that vanishing to leave a quantitative trace among the finite wedges.

9 New proof targets

This section turns the preceding accounting identities into explicit research problems. They are ordered by structural plausibility, not by estimated difficulty.

9.1 Target A: adelic defect transfer

For arbitrary depth define the exact local Smith credit

$$e_p(k) := 2k - ((k - \alpha_p)_+ + (k - \gamma_p)_+) = \min(k, \alpha_p) + \min(k, \gamma_p). \quad (79)$$

It is the exponent saved relative to the generic index p^{2k} .

Conjecture 9.1 (Macroscopic defect transfer). *Let integers $M, A > 1$ satisfy $D_\infty = 0$ but assume there is no rational q with $M = 2^q$ and $A = 3^q$. Then there exist finite sets S_j of good primes and depths $k_{p,j}$ along which*

$$\sum_{p \in S_j} e_p(k_{p,j}) \log p \geq (1 - o(1)) \sum_{p \in S_j} k_{p,j} \log p. \quad (80)$$

The truncation in (79) is exact: Smith credit beyond the requested depth cannot reduce the index further. A version strong enough to imply (80) would bring the exponent-lattice index down to almost linear scale by [Theorem 7.3](#); [Section 11](#) explains the additional direction compression still required for a contradiction.

The conjecture is intentionally stated as a target, not asserted as a known fact. It resembles a four-logarithm or adelic linear-independence principle, but its quantitative form is tailored to the exact amount of credit required. A weaker statement such as “ $\Delta_p = 0$ for infinitely many p ” is not enough: infinitely many isolated first-layer zeros can still have zero density and bounded total credit.

9.2 Target B: projective coherence of rank-one places

Suppose $D_p = 0$ and neither row of B_p vanishes. Then the two logarithm vectors are proportional in $\mathbb{Q}_p^2$. There is a projective slope

$$c_p = \frac{\log_p M}{\log_p 2} = \frac{\log_p A}{\log_p 3} \in \mathbb{Q}_p \quad (81)$$

whenever the denominators are nonzero. At the real place the analogous slope is β .

A collection of rank-one primes is useful only if their one-dimensional congruence directions are compatible in exponent space. This motivates:

Problem 9.2 (Moving-prime projective rigidity). *Show that if the slopes c_p exist for a sufficiently rich set of primes and the associated rank-one kernels admit exponent vectors of subquadratic combined index, then the common real slope is rational, hence integral.*

One possible route is to construct rational approximants to the adelic projective point $(c_v)_v$ and invoke a Subspace-Theorem statement on $\mathbf{P}^1$ with moving local targets. Unlike a direct application to the gaps, this formulation retains the rank information that the Smith law says is decisive.

9.3 Target C: a second-order, not first-order, Fermat law

The congruence $\Delta_p = 0$ records only $v_p(D_p) \geq 3$ in the unnormalized determinant. A proof needs information about the entire tower. For $u \in \mathbb{Z}_p^\times$ define higher logarithmic digits

$$\lambda_{p,j}(u) := p^{-j} \left(\log_p u - \sum_{i=1}^{j-1} p^i \lambda_{p,i}(u) \right) \pmod{p}. \quad (82)$$

Equivalently, one may use Witt coordinates of the principal unit $\langle u \rangle_p$. Expanding D_p digit by digit gives a hierarchy

$$\Delta_p^{(1)}, \Delta_p^{(2)}, \dots \quad (83)$$

whose first term is (53) and whose consecutive vanishing is equivalent to increasing τ_p .

Problem 9.3 (Higher-layer correlation). *Derive a global relation among the sequences $\lambda_{p,j}(2), \lambda_{p,j}(3), \lambda_{p,j}(M), \lambda_{p,j}(N)$ that uses the real identity $D_\infty = 0$ and forces either $D_p = 0$ or a large aggregate value of τ_p over a moving set of primes.*

This is a more accurate formulation than seeking many Wieferich-type primes. The objective is not first-digit vanishing but enough total Smith credit to change the lattice-cost exponent.

9.4 Target D: an auxiliary wedge with a product formula

The obstacle to a direct adelic argument is that D_v is not the logarithm or valuation of one algebraic number. A potentially overlooked approach is to manufacture an algebraic auxiliary quantity whose first nonzero local term is D_v .

For small formal parameters X, Y , consider the commutator-like expression

$$\Phi_v(X, Y) := (2^X M^Y - 1)(3^Y A^X - 1) - (2^Y M^X - 1)(3^X A^Y - 1). \quad (84)$$

Its lowest-order terms are alternating combinations of the two logarithm vectors. The raw expression in (84) is formal at present; the research goal is to discretize X, Y with integer interpolation or finite differences so that:

1. the resulting quantity is algebraic, preferably rational or integral;
2. its archimedean size is unusually small because $D_\infty = 0$;
3. its p -adic valuation detects τ_p ; and
4. the global product formula forces a nontrivial finite-place defect budget.

This is not the usual interpolation determinant built from a single exponential sequence. The alternating construction is designed to retain the two-by-two wedge that the local Smith analysis identifies.

9.5 Target E: exploit the global gcd theorem contrapositive

Theorem 4.5 can be used as a contradiction engine if one constructs synchronized exponents with a forced modulus N satisfying

$$\log N \geq c(r + s) \tag{85}$$

for some fixed $c > 0$. Generic congruence lattices cannot do this, but a rank-collapsed family might. The exact target is therefore:

Criterion 9.4 (A sufficient local construction). *A contradiction follows if there exist exponent pairs (r_j, s_j) and good-prime moduli N_j such that*

$$N_j \mid D_2(r_j, s_j), \quad N_j \mid D_3(r_j, s_j), \quad \liminf_{j \rightarrow \infty} \frac{\log N_j}{r_j + s_j} > 0. \tag{86}$$

The criterion is immediate from Theorem 4.5, but it focuses the search: local progress matters only insofar as it approaches an exponential divisor at linear exponent height. Tables of isolated local zeros should be evaluated against this threshold.

10 Exact computational reconnaissance

A finite computation cannot test a nonexistent counterexample pair (M, A) . It can, however, audit the local formulas and calibrate how special first-layer rank loss appears among nearby integer pairs. The following experiment scans the exact Fermat-quotient determinant (53); it uses modular exponentiation modulo p^2 , so no floating-point arithmetic enters.

10.1 Design of the scan

The pair $(M, A) = (4, 9)$ is an integral control corresponding to $\beta = 2$. For it, the real and every p -adic logarithm vector are exactly proportional, hence $\Delta_p = 0$ at every good prime. The other pairs are rough “nearest-output” proxies: A was chosen loosely near the power-law curve suggested by the real relation, only to sample arithmetic behavior, not because the pairs satisfy $\log M / \log 2 = \log A / \log 3$. No statistical inference about a genuine counterexample is made.

To avoid any ambiguity in that informal description, the computation is best regarded simply as a fixed list of integer test pairs. Its purposes are:

- verify the exact control identity;
- locate accidental first-layer zeros $\Delta_p = 0$;

- illustrate that a zero at one prime is not the same as rank one in $\mathbb{Q}_p$; and
- provide reproducible input for future higher-layer scans.

M	A	good primes	first nonzero Δ_p	zero-defect primes below 1000	role
4	9	166	–	all	integral control
5	13	164	7	none	proxy
7	25	164	11	163	proxy
11	41	164	5	7, 181	proxy
13	55	163	7	none	proxy
17	91	163	5	none	proxy
19	119	163	5	none	proxy
23	151	164	5	31, 251	proxy

Table 2: Exact first-layer determinant scan. “Good” means $p \geq 5$ and $p \nmid 6MA$. The integral control vanishes identically; sporadic zeros for proxy pairs are only first-layer events.

The control row is a useful implementation test: for $M = 4$ and $A = 9$, $q_p(4) = 2q_p(2)$ and $q_p(9) = 2q_p(3)$ modulo p , so $\Delta_p = 0$ identically. In the proxy rows, nonzero determinants appear immediately or very early, while occasional isolated zeros occur. The proper conclusion is modest: generic full rank is common in unrelated integer data, but first-layer accidents are not rare enough to carry structural meaning by themselves. A serious scan must compute $\tau_p = v_p(\det B_p)$ or certify $D_p = 0$ in $\mathbb{Q}_p$, rather than stop at Δ_p .

10.2 Reproducibility

The complete script is included in [Section D](#). Its central line is

$$q_p(u) = \frac{u^{p-1} - 1}{p} \pmod{p},$$

implemented by evaluating u^{p-1} modulo p^2 . The table can therefore be reproduced with standard Python alone.

11 The second scale mismatch: index reduction is not yet compression

The defect budget identifies when the local index drops from G^2 toward G . That is an important structural threshold, but it is not by itself large enough to contradict the global gcd theorem. The distinction is best seen by writing

$$L := \log G. \tag{87}$$

Then the generic index is e^{2L} and the rank-one index is e^L . Minkowski’s first theorem gives an unconstrained Euclidean short vector of size at most a constant times $I^{1/2}$, hence of scale e^L in the generic case and $e^{L/2}$ in the rank-one case. By contrast, a common divisor of scale $G = e^L$ contradicts [Theorem 4.5](#) only if the associated exponent height is $O(L)$, or at least if L/H stays bounded away from zero.

Thus there are two separate compressions:

1. *dimension compression*: reduce the congruence index from G^2 toward G by determinant defect or rank loss;
2. *direction compression*: produce a positive exponent vector of size $G^{o(1)}$, ideally $O(\log G)$, inside that lattice.

The first is measured by Smith invariants. The second is an archimedean orientation problem across all selected finite places. A successful proof needs both, or a different mechanism that bypasses the congruence lattice entirely.

11.1 Positive minimum of a congruence lattice

For a full-rank lattice $C \subset \mathbb{Z}^2$, define its positive minimum

$$\lambda_+(C) := \min \{r + s : (r, s) \in C, r \geq 1, s \geq 1\}. \quad (88)$$

The minimum exists because a finite-index subgroup of $\mathbb{Z}^2$ contains a positive multiple of each standard basis vector.

For a finite set of good primes with full depths $n_p \geq 1$, let

$$C_S(\mathbf{n}) := \bigcap_{p \in S} \{(r, s) : M^s \equiv 2^r \pmod{p^{n_p}}, A^s \equiv 3^r \pmod{p^{n_p}}\} \quad (89)$$

and

$$N_S(\mathbf{n}) := \prod_{p \in S} p^{n_p}. \quad (90)$$

Corollary 11.1 (No logarithmically short positive codeword). *Under the hypothetical counterexample (1), for any sequence $(S_j, \mathbf{n}_j)$ with $N_j := N_{S_j}(\mathbf{n}_j) \rightarrow \infty$,*

$$\frac{\log N_j}{\lambda_+(C_{S_j}(\mathbf{n}_j))} \rightarrow 0. \quad (91)$$

Proof. Choose a positive minimizing vector (r_j, s_j) . Then N_j divides both $D_2(r_j, s_j)$ and $D_3(r_j, s_j)$. The heights $r_j + s_j$ must tend to infinity: otherwise a bounded subsequence would contain a fixed positive pair, while the nonzero fixed integer gcd of its two gaps could not be divisible by $N_j \rightarrow \infty$. Theorem 4.5 now gives

$$\log N_j \leq \log \gcd(D_2(r_j, s_j), D_3(r_j, s_j)) = o(r_j + s_j),$$

which is (91). □

This corollary recasts the global theorem as a minimum-distance law for a family of arithmetic lattice codes. Any local argument claiming a small positive exponent vector can be checked against it immediately.

11.2 Successive minima and the one-short-direction loophole

Let $\lambda_1(C) \leq \lambda_2(C)$ be Euclidean successive minima. In rank two, Minkowski's second theorem gives

$$\lambda_1(C)\lambda_2(C) \asymp [\mathbb{Z}^2 : C] \quad (92)$$

with absolute constants. Therefore a lattice of very large index may still contain one exceptionally short vector; the compensating second vector is then very long. Determinant calculations alone cannot exclude this loophole.

For the Alaoglu–Erdős problem, a single short vector shared by many deep local congruences would already be striking. If the same fixed (r, s) survived arbitrarily large depths at one prime, then $M^s = 2^r$ and $A^s = 3^r$ as rational numbers, contradicting irrationality. The more subtle possibility is a moving sequence of short directions. A new proof could attempt to show that the real rank relation (76) prevents these directions from moving freely.

Problem 11.2 (Cross-prime direction coherence). *Prove an upper bound*

$$\lambda_+(C_S(\mathbf{n})) \leq (\log N_S(\mathbf{n}))^{O(1)} \quad (93)$$

for an unbounded sequence of moduli forced by $D_\infty = 0$, or prove a weaker bound that still makes $\log N_S/\lambda_+$ stay positive. By Theorem 11.1, either conclusion rules out a counterexample.

The problem is deliberately strong. It states the exact compression needed, exposing why a mere density theorem for primes with $\Delta_p = 0$ would not finish the argument.

11.3 Dual formulation

Let $C^* := \{y \in \mathbb{R}^2 : y \cdot C \subset \mathbb{Z}\}$ be the Euclidean dual lattice. The linear form $s\beta - r$ corresponds to the vector $(-1, \beta)$. A small value on C means that this real covector lies close, in a suitable strip norm, to a rational covector from C^* . Locally, the rows of $p^{-k}B_p$ generate the relevant modular dual constraints. Hence the missing adelic statement can also be phrased as:

Problem 11.3 (Dual adelic approximation). *Use $D_\infty = 0$ to show that the archimedean covector $(-1, \beta)$ is unusually close to a dual vector assembled from the local logarithm rows, with total denominator only polynomial in the requested depths rather than exponential in their modulus.*

This dual viewpoint may be better suited to an adelic transference theorem or to a geometry-of-numbers argument on an S -arithmetic lattice. It also suggests that the correct object is not a collection of unrelated local kernels, but one global dual class whose local realizations are the rows of B_p .

12 Additional consequences of synchronized-gap rigidity

The global theorem has several useful corollaries beyond the quotient height calculation.

12.1 Weighted common valuation is negligible

For every sequence $r + s \rightarrow \infty$,

$$\sum_p \min\{v_p(D_2), v_p(D_3)\} \log p = o(r + s). \quad (94)$$

Thus for any fixed $\eta > 0$, the set of primes satisfying

$$\min\{v_p(D_2), v_p(D_3)\} \log p \geq \eta(r + s) \quad (95)$$

is empty for all sufficiently large (r, s) . More generally, no bounded number of primes can carry a positive proportion of exponent height in common valuation.

12.2 Exponential coprime remainders

Along $r/s \rightarrow \beta$, the individual gaps have logarithmic sizes

$$\log |D_2(r, s)| = r \log 2 + o(s), \quad (96)$$

$$\log |D_3(r, s)| = r \log 3 + o(s), \quad (97)$$

provided $\delta_{r,s} = e^{-o(s)}$; this includes every sequence by [Theorem 2.2](#). Dividing by their gcd leaves

$$\log \frac{|D_2|}{\gcd(D_2, D_3)} = r \log 2 + o(s), \quad (98)$$

$$\log \frac{|D_3|}{\gcd(D_2, D_3)} = r \log 3 + o(s). \quad (99)$$

Hence almost all exponential mass of each gap is carried by prime powers not shared with the other gap. Any prospective argument based on the same prime support must therefore exploit unweighted support or residue correlations, not a large common numerical factor.

12.3 Fixed finite sets cannot dominate

Fix a finite set of primes S . Standard p -adic logarithmic-form bounds give

$$\sum_{p \in S} \min\{v_p(D_2), v_p(D_3)\} \log p = O_S((\log(r + s))^2). \quad (100)$$

Thus every substantial common divisor must move to new primes. This is a precise moving-prime requirement: neither a single favorable prime nor a fixed finite collection can generate the needed exponential scale.

12.4 An entropy interpretation

Let

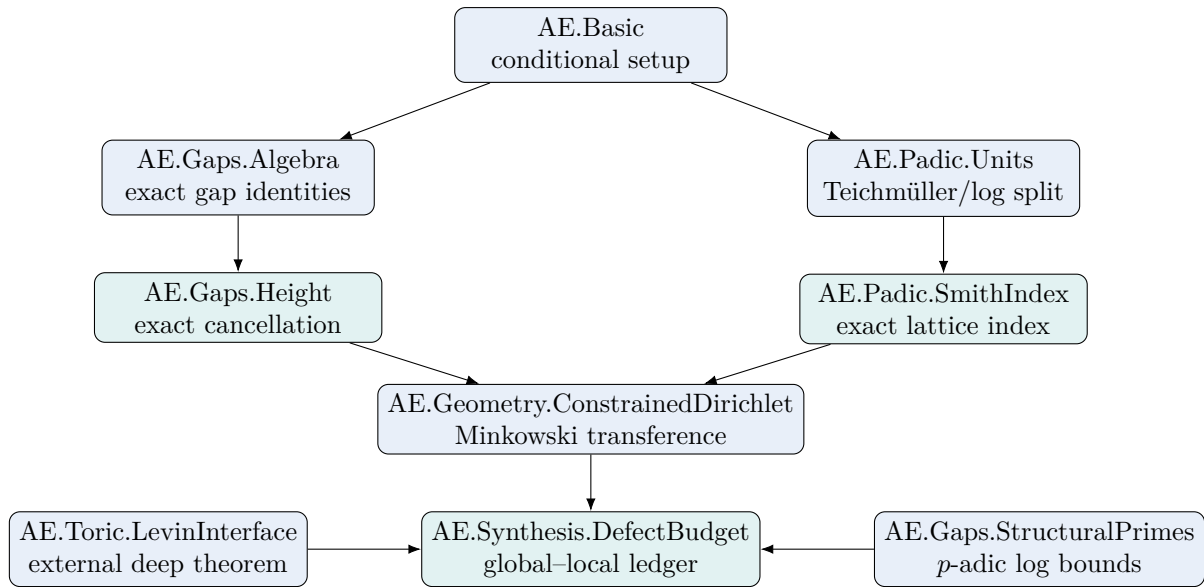
$$\mu_{r,s}(p) := \frac{\min\{v_p(D_2), v_p(D_3)\} \log p}{r + s}. \quad (101)$$

Then (94) says that the total mass $\sum_p \mu_{r,s}(p)$ tends to zero. Local lifting attempts to place mass at selected primes, while the global theorem says the total common mass vanishes. The only possible escape is to use local data that does not manifest as a common divisor—for example, correlated projective slopes, local determinant defects, or auxiliary quantities with cancellation before taking valuations. This helps explain why the adelic wedge (78) is a more promising carrier of information than the raw common factor alone.

13 A formal proof-development plan in Lean

The new elementary and algebraic layers are well suited to formalization. The deep toric generalized-gcd input is not currently a routine Mathlib component, so a credible formal project should isolate it behind a clearly named interface rather than silently axiomatize the final conjecture.

13.1 Module graph



13.2 Core theorem signatures

The following sketches use Mathlib-style names but are intentionally not advertised as compiling code. They specify the interfaces that a formal implementation should realize.

Listing 1: Proposed Lean interfaces for the elementary core.

```

-- AE/Basic.lean
structure HypotheticalCounterexample where
  beta : Real
  M A : Nat
  hM : Real.rpow 2 beta = M
  hA : Real.rpow 3 beta = A
  hbeta : beta \notin Set.range ((Int.cast : Int -> Real))

-- AE/Padic/SmithIndex.lean
-- B is a 2 x 2 matrix over Z_p; smithExponents records alpha <= gamma.
theorem index_preimage_smith_rank_two
  (B : Matrix (Fin 2) (Fin 2) Zp)
  (hsmith : SmithData B alpha gamma)
  (k : Nat) :
  Fintype.card (ZModLatticeQuotient B k)
    = p ^ ((k - alpha) + (k - gamma)) := by
  ...

-- AE/Geometry/ConstrainedDirichlet.lean
theorem exists_lattice_approx
  (Lambda : Submodule Z (Fin 2 -> Z))
  (hfinite : FiniteIndex Lambda)
  (beta : Real) (hbeta : Irrational beta)
  (Q : Real) (hQ : 1 < Q) (eta : Real) (heta : 0 < eta) :
  exists r s : Z,
    vector![r,s] \in Lambda /\ 0 < s /\
    s <= (1+eta) * index Lambda * Q /\
    abs (s*beta-r) <= Q^(-1) := by
  ...

-- AE/Gaps/Height.lean
theorem reduced_gcd_eq_prime_to_six

```

```

(h : HypotheticalCounterexample) (r s : Nat)
(hr : valuation 2 h.M * s < r)
(hs : valuation 3 h.A * s < r) :
reducedCommonFactor (gap2 h r s) (gap3 h r s)
  = gcdPrimeToSix (gap2 h r s) (gap3 h r s) := by
...

```

The pseudo-code uses subtraction on natural numbers only schematically; a real implementation should use truncated subtraction explicitly or state the stable-depth theorem with hypotheses $k \geq \gamma$.

13.3 Formalization order

A sensible order minimizes dependence on difficult analysis:

1. formalize the integer factorization identities (26) and (27);
2. formalize finite abelian-group indices and the coprime-intersection lemma used in [Theorem 6.6](#);
3. prove the Smith-index law first for diagonal matrices over $\mathbb{Z}/p^n\mathbb{Z}$, then transport through invertible row and column operations;
4. formalize the determinant-one map T_β and invoke the existing geometry-of-numbers infrastructure for Minkowski;
5. isolate asymptotic facts about $o(s)$ and height in a separate file, with Baker’s lower bound supplied as an imported theorem;
6. represent Levin’s theorem as a precisely scoped interface whose hypotheses include coprimality, finite generation, and the exceptional-subtorus conclusion;
7. formalize the rank-one exceptional descent independently of the proof of Levin’s theorem.

The last item is especially valuable. Even before the deep theorem is in Mathlib, one can machine-check that its exceptional set is eliminated correctly in this specific rank-two orbit. That is the most delicate new logical step in [Theorem 4.2](#).

13.4 What should not be hidden behind an axiom

A formal artifact should not introduce an axiom equivalent to “the synchronized gcd is subexponential.” It should import only the general toric gcd theorem and prove the following in Lean:

- injectivity and Zariski density of the exponent orbit;
- rank at most one for intersections with proper subtori;
- multiplicative independence of every cyclic direction;
- finiteness of second-stage exceptional intersections;
- exact identification of good-prime local gcd terms; and
- reduction of the primes 2, 3 to fixed-base p -adic logarithmic forms.

This separation makes the formalization scientifically informative rather than ceremonial.

14 Limitations, verification boundaries, and next experiment

The report has intentionally separated theorem, conditional theorem, conjectural target, and computation. The boundaries are as follows.

1. **No complete proof.** The new results do not yet force $\beta \in \mathbb{Z}$. They identify a sharper missing theorem.
2. **Deep external input.** The global gcd theorem uses Levin’s generalized-gcd theorem on algebraic tori. The exceptional-line descent is proved here, but the underlying toric theorem is not reproved.
3. **Effectivity.** The $o(r + s)$ estimate inherits the ineffectivity or poor effectivity of Subspace-Theorem technology. It supplies a structural asymptotic obstruction, not a practical numerical cutoff.
4. **Structural primes.** The primes 2 and 3 are controlled by standard p -adic logarithmic-form estimates. Their precise constants are irrelevant to the asymptotic theorem and are not optimized.
5. **Proxy computation.** The pairs in Table 2 do not satisfy the counterexample relation, except for the integral control. The scan validates formulas and illustrates first-layer behavior; it is not statistical evidence for or against the conjecture.
6. **Adelic targets.** The defect-transfer and direction- coherence statements are proposed research targets. They are not consequences of known four-logarithm theorems as presently stated.

The most informative next computational experiment is not a wider scan of Δ_p . It is a depth scan for the exact Smith data. For each test pair and prime, compute B_p to precision p^K , determine α_p, γ_p , and compare the observed index of congruence solutions modulo p^k with (47). The output should record

$$(p, \alpha_p, \gamma_p, \tau_p, t_p, \lambda_+(C_p(k)))_{1 \leq k \leq K}. \quad (102)$$

That experiment tests not only determinant depth but also the movement of short exponent directions, the second compression identified in Section 11.

15 Conclusion

The central advance is a global rigidity law:

$$\log \gcd(M^s - 2^r, A^s - 3^r) = o(r + s).$$

It says that synchronized gaps cannot share exponential mass. The proof uses the rank-two torus orbit, then explicitly descends through the finite exceptional union by showing that every exceptional exponent line carries a Zariski-dense cyclic direction. This closes the usual exceptional-set loophole.

The exact quotient-height theorem then shows that dividing the two gaps does not create powerful rational approximations to $\log_2 3$: after all cancellation, the height remains $\exp\{s \log(M_0 A_0) + o(s)\}$, while the error is only polynomially small in s .

At finite places, the Smith law gives a complete local ledger. Generic additional precision G costs index G^2 ; determinant defects contribute an exact credit E ; true rank loss changes the slope. Minkowski’s theorem turns the index into an archimedean error and exposes the square-root tax.

A further comparison with the global gcd theorem reveals a second, more severe mismatch: even an index of order G does not automatically produce positive exponent vectors of order $\log G$. Cross-prime direction coherence is therefore as important as rank loss.

The most promising unexplored route is an adelic wedge principle that transfers the real determinant identity $D_\infty = 0$ into both macroscopic finite-place defect and coherent short directions. The exact thresholds are now stated in (71), (93), and Theorem 9.4. These replace the open-ended instruction “find more favorable primes” with quantitative success criteria.

A Detailed exceptional-subtorus descent

This appendix expands the uniformity argument in Theorem 4.2, since that is the point most likely to be underestimated in an informal proof.

A.1 Characters and exponent kernels

Every connected one-dimensional subtorus $H \subset \mathbb{G}_m^2$ is the kernel of a primitive character

$$\chi_{u,v}(X, Y) = X^u Y^v, \quad \gcd(u, v) = 1. \quad (103)$$

A translate $C = \xi H$ is cut out by

$$X^u Y^v = c \quad (104)$$

for a nonzero algebraic constant c . Pulling back along $P_{r,s}$ gives

$$2^{-ur} 3^{-vr} M^{us} A^{vs} = c. \quad (105)$$

If two exponent pairs (r, s) and (r', s') solve the same equation, their difference $(R, S) = (r - r', s - s')$ satisfies

$$2^{-uR} 3^{-vR} M^{uS} A^{vS} = 1. \quad (106)$$

Thus the difference set is a subgroup of $\mathbb{Z}^2$. It cannot have rank two, for then Γ would have a finite-index subgroup contained in H and its Zariski closure would be proper, contradicting Theorem 3.1. Hence it has rank at most one, proving the affine-line description without invoking a general structure theorem for intersections of finitely generated groups with cosets.

A.2 A cyclic orbit meets a proper coset at most once after quotient

Let $q = (\rho, \sigma)$ have multiplicatively independent coordinates and consider $x_t = cq^t$. If a proper torus coset ξH contains both x_{t_1} and x_{t_2} , then

$$q^{t_1 - t_2} \in H. \quad (107)$$

If H is defined by $X^u Y^v = 1$, this gives $(\rho^u \sigma^v)^{t_1 - t_2} = 1$. Since ρ, σ are positive rational numbers in the present application, the only root of unity they can produce is 1, so $\rho^u \sigma^v = 1$. Multiplicative independence forces $u = v = 0$, contradicting properness of H . Thus each proper coset contains at most one term of the cyclic orbit. For a finite union, the intersection is finite with an explicit cardinality bounded by the number of cosets.

A.3 Uniform little- o across finitely many lines

Fix a target $\varepsilon > 0$. On the generic part of the rank-two group, apply Levin with parameter $\varepsilon/(3C_0)$, where

$$h(P_{r,s}) \leq C_0(r+s) + C_0. \quad (108)$$

This gives the desired estimate outside finitely many cosets for large $r+s$.

For each exceptional line L_i , write $(r, s) = (r_i, s_i) + t(a_i, b_i)$. There are constants $c_i, C_i > 0$ such that, outside finitely many t ,

$$c_i|t| - C_i \leq r + s \leq C_i|t| + C_i \quad (109)$$

when restricted to the positive quadrant. Apply Levin to the cyclic group on L_i with parameter $\varepsilon c_i/(3C'_i)$, where C'_i is a height constant for $(c_1\rho^t, c_2\sigma^t)$. After excluding the finite second-stage intersections, the local gcd is at most $\varepsilon(r+s)/3$ for all large positive-quadrant points on L_i .

The number of lines is finite. Taking the maximum of their thresholds, and absorbing all isolated and initially excluded points into one finite set, proves a single $H_0(\varepsilon)$ valid on the entire orbit. This is the quantifier needed for (17).

B Smith normal form and index details

B.1 One-dimensional calculation

Let $e, k \geq 0$. The subgroup

$$L_{e,k} := \{x \in \mathbb{Z}_p : p^e x \in p^k \mathbb{Z}_p\} \quad (110)$$

is

$$L_{e,k} = p^{(k-e)_+} \mathbb{Z}_p, \quad [\mathbb{Z}_p : L_{e,k}] = p^{(k-e)_+}. \quad (111)$$

The two-dimensional formula is simply the product of two copies of (111) after Smith reduction.

B.2 Why integral and p -adic indices agree

Let

$$\Lambda = \{z \in \mathbb{Z}^2 : Bz \in p^k \mathbb{Z}_p^2\}. \quad (112)$$

The condition depends only on z modulo a sufficiently high power p^N , for example $N = k + \max(0, -\min v_p(B_{ij}))$ before normalization. Hence Λ is the inverse image of a subgroup of $(\mathbb{Z}/p^N \mathbb{Z})^2$. Its index equals the cardinality of the corresponding quotient. Completing at p produces the same finite quotient because $\mathbb{Z}^2$ is dense in $\mathbb{Z}_p^2$ and reduction modulo p^N is surjective. Thus no subtlety is hidden in passing from $\mathbb{Z}^2$ to $\mathbb{Z}_p^2$ in Theorem 6.2.

B.3 Smith invariants from minors

For a rank-two matrix $B \in M_2(\mathbb{Z}_p)$,

$$\alpha = \min_{i,j} v_p(B_{ij}), \quad \alpha + \gamma = v_p(\det B). \quad (113)$$

Therefore a computation of $\tau = v_p(\det B)$ alone does not determine the onset depth γ unless the minimum entry valuation α is also known. In the common generic situation where at least one normalized logarithm is a p -adic unit, $\alpha = 0$ and $\gamma = \tau$. Then

$$[\mathbb{Z}^2 : \Lambda_p(k)] = p^{k+(k-\tau)_+}. \quad (114)$$

The index initially grows with slope one until depth τ , then with slope two. This finite initial interval is precisely what a first-layer Fermat zero detects; it should not be mistaken for permanent rank one.

B.4 Torsion-principal separation

The map

$$\varphi_p : \mathbb{Z}^2 \rightarrow (\mathbb{F}_p^\times)^2, \quad (r, s) \mapsto (M^s 2^{-r}, A^s 3^{-r}) \quad (115)$$

has image of cardinality t_p and kernel T_p . The principal-unit map at depth k has a p -group image. Since one image has order prime to p and the other has p -power order, the combined map has image the direct product of the two images. This gives a second proof of (60) and is a convenient formulation for Lean.

C Height calculation details

C.1 Uniform control of $b^\delta - 1$

Let $\delta = o(s)$ be nonzero and satisfy $|\delta| \geq s^{-C}$. For fixed $b > 1$:

- if $|\delta| \leq 1$, the mean-value theorem gives $|b^\delta - 1| \asymp_b |\delta|$, so its logarithm lies between $-C \log s + O(1)$ and $O(1)$;
- if $\delta > 1$, then $0 < \log(b^\delta - 1) \leq \delta \log b = o(s)$;
- if $\delta < -1$, then $|b^\delta - 1|$ is bounded above and below by positive constants.

Hence $\log |b^\delta - 1| = o(s)$ uniformly in all three regimes. This is the only analytic input needed after Baker’s lower bound.

C.2 Exact cancellation prime by prime

For $p \geq 5$,

$$\begin{aligned} v_p(2^{r-as} E_3) &= v_p(E_3) = v_p(D_3), \\ v_p(3^{r-bs} E_2) &= v_p(E_2) = v_p(D_2). \end{aligned}$$

At $p = 2$, the denominator is odd; at $p = 3$, the numerator is prime to 3. Therefore the reduced common factor is exactly

$$\prod_{p \geq 5} p^{\min(v_p(D_2), v_p(D_3))},$$

with no hidden cancellation at the structural primes. This exactness is what allows the global gcd theorem to feed directly into the height asymptotic.

D Reproducible Fermat-quotient scan

Listing 2: Exact script used for Table 2.

```
#!/usr/bin/env python3
"""Exact first-layer p-adic determinant reconnaissance."""
from __future__ import annotations
```

```

from math import gcd

def primes_below(n: int) -> list[int]:
    sieve = bytearray(b"\x01" * n)
    sieve[:2] = b"\x00\x00"
    for p in range(2, int(n**0.5) + 1):
        if sieve[p]:
            sieve[p*p:n:p] = b"\x00" * ((n-1-p*p)//p)+1
    return [i for i in range(2, n) if sieve[i]]

def fermtat_quotient(u: int, p: int) -> int:
    # Reducing modulo p^2 keeps the quotient exact.
    return ((pow(u, p - 1, p * p) - 1) // p) % p

def defect(M: int, A: int, p: int) -> int:
    qM = fermtat_quotient(M, p)
    qA = fermtat_quotient(A, p)
    q2 = fermtat_quotient(2, p)
    q3 = fermtat_quotient(3, p)
    return (qM * q3 - qA * q2) % p

def summarize(M: int, A: int, bound: int = 1000):
    good = [p for p in primes_below(bound)
            if p >= 5 and gcd(p, 6*M*A) == 1]
    zeros = [p for p in good if defect(M, A, p) == 0]
    first_nonzero = next(
        (p for p in good if defect(M, A, p) != 0), None)
    return len(good), zeros, first_nonzero

def main() -> None:
    pairs = [
        (4, 9, "integral control, beta=2"),
        (5, 13, "proxy"),
        (7, 25, "proxy"),
        (11, 41, "proxy"),
        (13, 55, "proxy"),
        (17, 91, "proxy"),
        (19, 119, "proxy"),
        (23, 151, "proxy"),
    ]
    for M, A, description in pairs:
        total, zeros, first = summarize(M, A)
        print(M, A, description, total, zeros, first)

if __name__ == "__main__":
    main()

```

E Checklist for independent verification

An independent reader can audit the new chain in the following order.

1. Verify irrationality of a nonintegral hypothetical β and the Zariski density of Γ .
2. Check the statement of Levin's theorem for $X - 1, Y - 1$ on a finitely generated subgroup of

$\mathbb{G}_m^2$.

3. Verify that each first-stage exceptional coset pulls back to rank at most one in exponent space.
4. Check the calculation $(b\beta - a)(u \log 2 + v \log 3) = 0$ on every exceptional line.
5. Apply the toric theorem a second time and verify finite intersection with the new exceptional cosets.
6. Check the $p = 2, 3$ reductions and invoke any standard fixed-base p -adic logarithmic-form bound.
7. Verify the exact quotient factorization and cancellation at 2, 3.
8. Check the Smith diagonal preimage one coordinate at a time.
9. Verify the coprimality of torsion and principal indices.
10. Apply Minkowski to the rectangle with area $4(1 + \eta)I$.

Every step except items 2 and 6 is elementary or algebraic. Items 2 and 6 are standard external theorems with explicit references below.

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